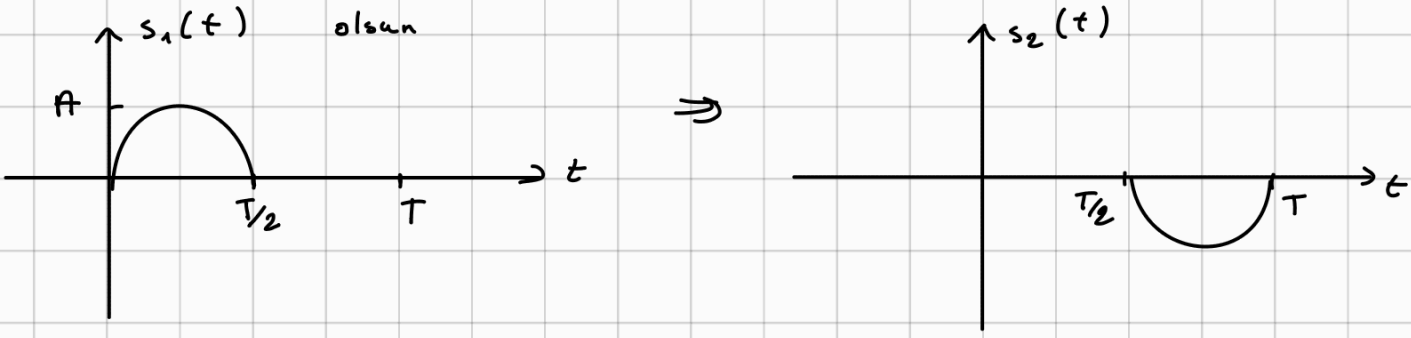
$$s_1(t) = \begin{cases} A \sin\left(\frac{2\pi t}{T}\right), & 0 \le t \le T/2 \\ 0, & \text{Else} \end{cases}$$

$$s_2(t) = -s_1(t - T/2)$$



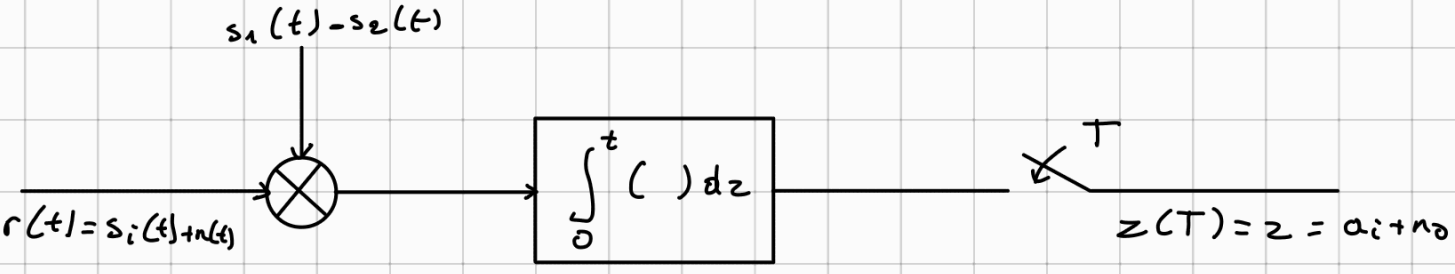
* $s_1(t)$ signal is transmitted for the bit "1"
* $s_2(t)$ signal is transmitted for the bit "0"

Find the bit error (BER) expression of this system over additive white gaussian channel (AWGN) for

a) $P(1) = P(0) = 1/2$

b) $P(1) = 1/4$, $P(0) = 3/4$

and plot it.



$$a_i = a_i(T) = \int_0^T s_i(t)\,[s_1(t) - s_2(t)]\,dt$$

$$a_1(T) = \int_0^T s_1(t)\,[s_1(t) - s_2(t)]\,dt = \int_0^{T/2} s_1^2(t) - \int_0^T s_1(t)\,s_2(t)\,dt$$

$$a_1(T) = \int_0^{T/2} A^2 \sin^2\left(\frac{2\pi t}{T}\right) dt$$

$$= \frac{A^2}{2} \int_0^{T/2} 1 - \cos\left(\frac{4\pi t}{T}\right) dt$$

$$\sin^2 x = \frac{1 - \cos(2x)}{2}$$

$$= \frac{A^2}{2}\left[t - \frac{\sin\left(\frac{4\pi t}{T}\right)}{\frac{4\pi}{T}} \right]_0^{T/2}$$

$$= \frac{A^2}{2}\left[t - T\,\frac{\sin\left(\frac{4\pi t}{T}\right)}{4\pi} \right]_0^{T/2}$$

$$= \frac{A^2}{2}\left[\left(\frac{T}{2} - 0\right) - \left[T\,\frac{\sin\left(\frac{4\pi}{T}\cdot\frac{T}{2}\right)}{4\pi} - \frac{T\sin(0)}{4\pi} \right] \right]$$

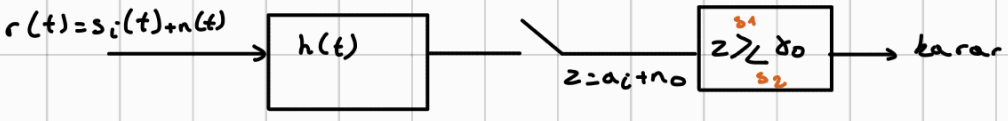
$$= \frac{A^2}{2}\left[\frac{T}{2} - \left[T\,\frac{\sin(2\pi)}{4\pi} - \frac{T\sin(0)}{4\pi} \right] \right]$$

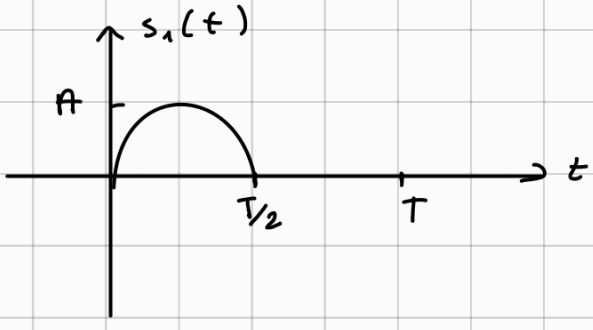
$$a_1(T) = \frac{A^2 T}{4}$$

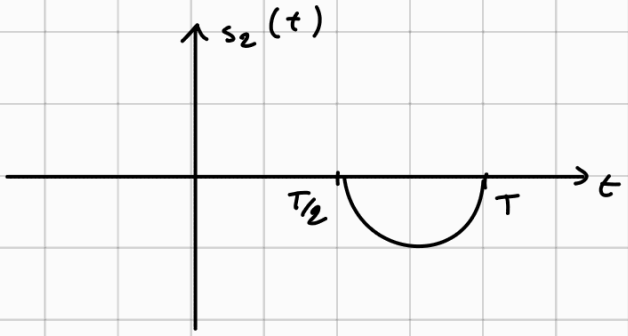
$$a_2(T) = \int_0^T s_2(t)\left[s_1(t) - s_2(t) \right] dt$$

$$a_2(T) = \int_0^T s_1(t)\,s_2(t) - \int_0^T s_2^2(t)\,dt = -\int_{T/2}^{T} s_2^2(t)\,dt$$

$$= -\int_0^{T/2} s_1^2(t)\,dt = -a_1(T) = -\frac{A^2 T}{4}$$







$$E_{s_1} = \frac{A^2}{2}\cdot\frac{T}{2} = \frac{A^2 T}{4} \qquad E_{s_2} = \frac{A^2}{2}\cdot\frac{T}{2} = \frac{A^2 T}{4}$$

$$E_b = E_{s_1} P(s_1) + E_{s_2} P(s_2) \quad \text{and} \quad E_{s_1} = E_{s_2}$$

$$E_b = E_{s_1}\underbrace{\left(P(s_1) + P(s_2)\right)}_{1} = E_{s_1} = \frac{A^2 T}{4}$$

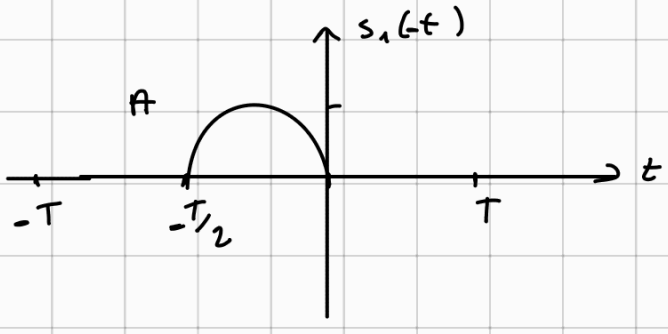
$a_1(T) = \frac{A^2 T}{4}$, $a_2(T) = -a_1(T)$ if $E_b = 1$ $a_1(T) = 1$, $a_2(T) = -1$

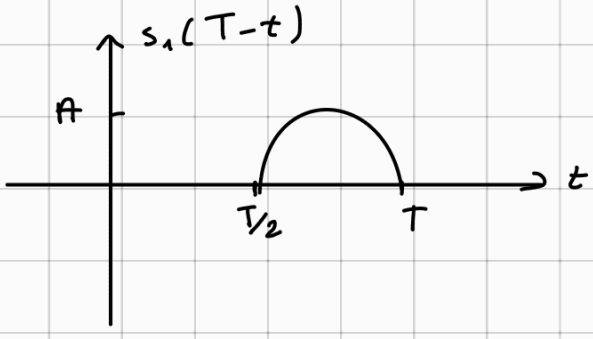
$\gamma_0 = \frac{a_1 + a_2}{2} = 0$ when $P(s_1) = P(s_2) = \frac{1}{2}$

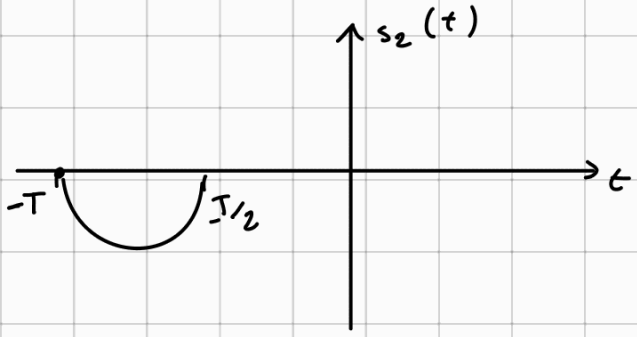
$\gamma_0 = \frac{\sigma_0^2}{a_1 - a_2} \ln \frac{P(s_2)}{P(s_1)} + \frac{a_1 + a_2}{2}$ general formula //

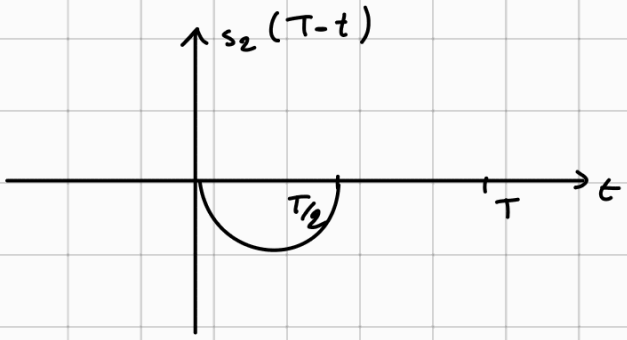
$\sigma_0^2 = \frac{N_0}{2} E_h$ so we have to find E_h

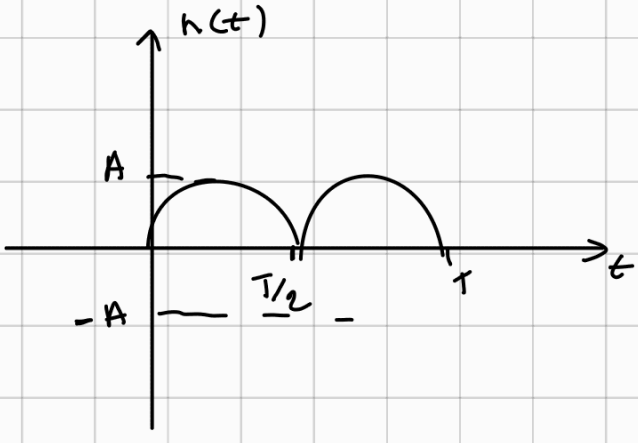
$h(t) = s_2(T-t) - s_2(T-t)$











$\Rightarrow E_h = \frac{A^2}{2} \cdot T = \frac{A^2 T}{2}$

$\Rightarrow \sigma_0^2 = \frac{N_0}{2} E_h = \frac{N_0 A^2 T}{4} \Rightarrow \sigma_0 = \frac{A}{2} \sqrt{N_0 T}$

if $P(s_1) = P(s_2) \Rightarrow P_b = Q\left(\frac{a_1 - a_2}{2\sigma_0}\right) = Q\left(\frac{\frac{A^2T}{4} - \left(-\frac{A^2T}{4}\right)}{2\frac{A}{2}\sqrt{N_0T}}\right) = Q\left(\frac{A^2T}{2A\sqrt{N_0T}}\right) =$

$= Q\left(\sqrt{\frac{A^2T}{4N_0}}\right) = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$

$$P(1) = P(0) = \frac{1}{2} \Rightarrow P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

special case

if $E_b = 1 \Rightarrow P_b = Q\left(\sqrt{\frac{1}{N_0}}\right)$

Then if $P(1) = \frac{1}{4} = P(s_1)$, $P(0) = \frac{3}{4} = P(s_2)$

$$\gamma_0 = \frac{\sigma_0^2}{a_1 - a_2} \ln\frac{P(s_2)}{P(s_1)} + \frac{a_1 + a_2}{2}$$

and $P_b = \left[1 - Q\left(\frac{\gamma_0 - a_1}{\sigma_0}\right)\right] P(s_1) + Q\left(\frac{\gamma_0 - a_2}{\sigma_0}\right) P(s_2)$

$$\Rightarrow \gamma_0 = \frac{\frac{N_0A^2T}{4}}{\frac{A^2T}{4} - \left(\frac{A^2T}{4}\right)} \ln\frac{\left(\frac{3}{4}\right)}{\frac{1}{4}} + \frac{\frac{A^2T}{4} - \frac{A^2T}{4}}{2} \to 0$$

$$\Rightarrow \gamma_0 = \frac{\frac{N_0A^2T}{4}}{\frac{A^2T}{2}} \ln(3) = \frac{N_0}{2}\ln(3) \approx (0.549)N_0 //$$

$$P_b = \left[1 - Q\left(\frac{\frac{N_0\ln(3)}{2} - \frac{A^2T}{4}}{\frac{A}{2}\sqrt{N_0T}}\right)\right]\frac{1}{4} + Q\left(\frac{\left(\frac{N_0\ln(3)}{2}\right) - \left(-\frac{A^2T}{4}\right)}{\frac{A}{2}\sqrt{N_0T}}\right)\cdot\frac{3}{4}$$

$$P_b = \left[1 - Q\left(\frac{\frac{N_0\ln(3)}{2} - \frac{A^2T}{4}}{\frac{A}{2}\sqrt{N_0T}}\right)\right]\frac{1}{4} + Q\left(\frac{\left(\frac{N_0\ln(3)}{2}\right) + \frac{A^2T}{4}}{\frac{A}{2}\sqrt{N_0T}}\right)\cdot\frac{3}{4}$$

special case

$E_b = \frac{A^2T}{4}$ if $E_b = 1$ $\frac{A^2T}{4} = 1 \Rightarrow \frac{A}{2}\sqrt{T} = 1$

$$\Rightarrow P_b = \left[1 - Q\left(\frac{\frac{N_0\ln(3)}{2} - 1}{\sqrt{N_0}}\right)\right]\frac{1}{4} + Q\left(\frac{\frac{N_0\ln 3}{2} + 1}{\sqrt{N_0}}\right)\frac{3}{4}$$

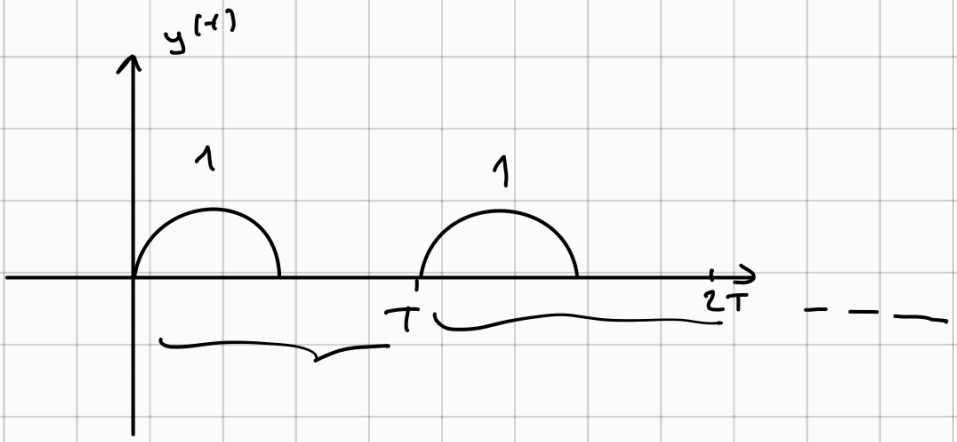
SNR$_{db}$ elimizdevar

$$SNR_{db} = 10 \log_{10} \frac{E_b}{N_0}$$

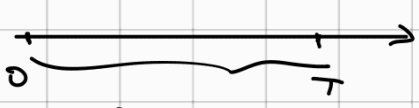
$$10^{\left(\frac{SNR_{db}}{10}\right)} = \frac{E_b}{N_0}$$

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

$$P_b = Q\left(\sqrt{2 \cdot 10^{\left(\frac{SNR_{db}}{10}\right)}}\right)$$



bit number

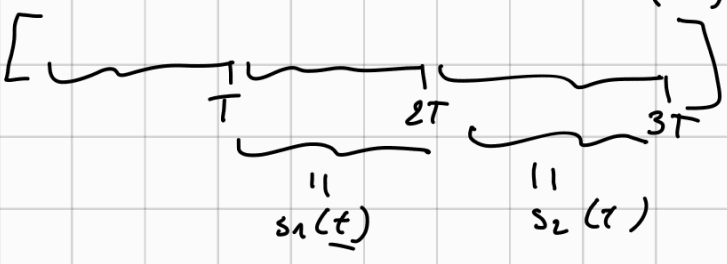


$t = 0 : T-1$

zaman = $\frac{T}{\text{length}(t)} \cdot$ bit number where length(t) = T iken

$$[\,t : T\,] \rightarrow \begin{matrix} 1 \\ 0 \end{matrix} = \begin{matrix} s_1(t) \\ s_2(t) \end{matrix}$$

$i.[1:T]$ → $(i-1)\,le.(t) < 1$ ∴ $leq < (t)$



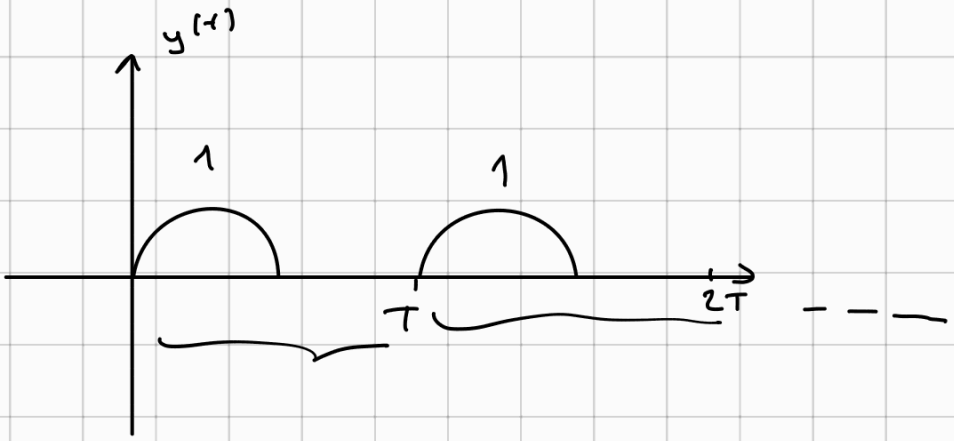
SNR$_{db}$ elimizdevar

$$SNR_{db} = 10 \log_{10} \frac{E_b}{N_0}$$

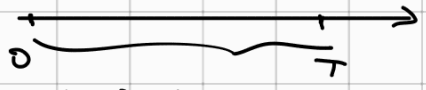
$$10^{\left(\frac{SNR_{db}}{10}\right)} = \frac{E_b}{N_0}$$

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

$$P_b = Q\left(\sqrt{2 \cdot 10^{\left(\frac{SNR_{db}}{10}\right)}}\right)$$



bit number



$t = 0 : T-1$

zaman = $\frac{T}{length(t)} \cdot bit\ number$ where $length(t) = T$ iken

$$[\ t : T\ ] \rightarrow \begin{matrix} 1 \\ 0 \end{matrix} = \begin{matrix} s_1(t) \\ s_2(t) \end{matrix}$$

$i.[1:T]$ → $(i-1)\,le.(t)_{<1}$, $i.\,leng<(t)$

